Cloud (AZ-900) · Quick concepts

Interview revision sheet. Read it the evening before, then test yourself in Interview practice.

Cloud concepts

- Consumption-based pricing; CapEx (upfront hardware) vs OpEx (pay as you go).
- Public, private and hybrid cloud.
- IaaS (VMs: you manage the OS), PaaS (App Service: you manage code), SaaS (Microsoft 365).
- Shared responsibility: the customer always owns data, accounts and devices.
- Benefits: high availability, scalability, elasticity, reliability, predictability, security, governance, manageability.

Azure architecture

- Region: area with data centres. Availability zones: at least three separate locations in a region.
- Region pairs for disaster recovery. Sovereign regions such as Azure Government.
- Hierarchy: management groups, subscriptions, resource groups, resources.
- A resource belongs to exactly one resource group.

Compute

- Virtual Machines (IaaS), VM Scale Sets (autoscale), availability sets (fault and update domains).
- App Service (PaaS web apps), Functions (serverless), Container Instances (simple containers), AKS (managed Kubernetes).
- Azure Virtual Desktop for cloud desktops.

Networking

- VNet and subnets; VNet peering connects VNets privately.
- VPN Gateway: encrypted over the internet. ExpressRoute: private dedicated connection.
- Azure DNS; private endpoints give PaaS services a private IP.

Storage

- Blob (objects), Files (SMB shares), Queues (messages), Tables, Disks.
- Access tiers: Hot, Cool, Cold, Archive (cheapest to store, slow and costly to read).
- Redundancy: LRS (one data centre), ZRS (zones), GRS (second region), GZRS; RA- adds read access to the secondary.
- Tools: AzCopy, Storage Explorer, Azure File Sync, Azure Migrate, Data Box.

Identity and security

- Microsoft Entra ID: cloud identity, SSO, MFA, passwordless, Conditional Access.
- External ID for partners (B2B). Azure RBAC: Owner, Contributor, Reader.
- Zero Trust: verify explicitly, least privilege, assume breach. Defense in depth.
- Microsoft Defender for Cloud: security posture and threat protection.

Cost and governance

- Pricing Calculator (estimate), TCO Calculator (on-prem vs Azure), Cost Management (budgets).
- Cost factors: resource type, usage, region, outbound data transfer; Reservations save money.
- Tags organise resources. Azure Policy enforces rules. Locks: CanNotDelete and ReadOnly.
- Microsoft Purview for data governance; Service Trust Portal for compliance reports.

Management and monitoring

- Portal, Cloud Shell, CLI, PowerShell; ARM templates and Bicep for IaC; Azure Arc for hybrid.
- Azure Advisor (recommendations), Service Health (Azure incidents), Azure Monitor (metrics, logs, alerts).
- Composite SLA of dependent services = product of their SLAs (lower than either).